

Analysis of a Quantum Cloner

1 Problem Statement

Consider a quantum cloner that works as follows: for any input qubit in the state

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \quad (1)$$

the cloner produces two qubits such that, with equal probabilities:

- The first qubit is in $|\psi\rangle$ and the second qubit in $|0\rangle$, OR
- The first qubit is in $|0\rangle$ and the second qubit in $|\psi\rangle$

We need to:

1. Compute the single-copy fidelity $F(\psi)$
2. Determine if the cloner is symmetric
3. Determine if the cloner is universal
4. Compute the average single-copy fidelity F_{average}
5. Compare with the Bužek-Hillery optimal symmetric universal quantum cloning machine

2 Step 1: Output State of the Cloner

The output state is a classical mixture:

$$\rho_{\text{out}} = \frac{1}{2} |\psi\rangle \langle\psi| \otimes |0\rangle \langle 0| + \frac{1}{2} |0\rangle \langle 0| \otimes |\psi\rangle \langle\psi| \quad (2)$$

3 Question 1: Single-Copy Fidelity

3.1 Definition

The single-copy fidelity is defined as the fidelity between the reduced state of one output qubit and the input state $|\psi\rangle$:

$$F(\psi) = \langle\psi| \rho_1 |\psi\rangle \quad (3)$$

where ρ_1 is the reduced density matrix of the first qubit (or second qubit).

3.2 Reduced Density Matrix of First Qubit

Tracing out the second qubit:

$$\rho_1 = \text{Tr}_2[\rho_{\text{out}}] \quad (4)$$

$$= \text{Tr}_2 \left[\frac{1}{2} |\psi\rangle \langle\psi| \otimes |0\rangle \langle 0| \right] + \text{Tr}_2 \left[\frac{1}{2} |0\rangle \langle 0| \otimes |\psi\rangle \langle\psi| \right] \quad (5)$$

$$= \frac{1}{2} |\psi\rangle \langle\psi| \cdot \text{Tr}[|0\rangle \langle 0|] + \frac{1}{2} |0\rangle \langle 0| \cdot \text{Tr}[|\psi\rangle \langle\psi|] \quad (6)$$

$$= \frac{1}{2} |\psi\rangle \langle\psi| \cdot 1 + \frac{1}{2} |0\rangle \langle 0| \cdot 1 \quad (7)$$

$$= \frac{1}{2} |\psi\rangle \langle\psi| + \frac{1}{2} |0\rangle \langle 0| \quad (8)$$

3.3 Computing the Fidelity

$$F(\psi) = \langle\psi| \rho_1 |\psi\rangle \quad (9)$$

$$= \langle\psi| \left(\frac{1}{2} |\psi\rangle \langle\psi| + \frac{1}{2} |0\rangle \langle 0| \right) |\psi\rangle \quad (10)$$

$$= \frac{1}{2} \langle\psi|\psi\rangle \langle\psi|\psi\rangle + \frac{1}{2} \langle\psi|0\rangle \langle 0|\psi\rangle \quad (11)$$

$$= \frac{1}{2} \cdot 1 + \frac{1}{2} |\langle\psi|0\rangle|^2 \quad (12)$$

Now compute $\langle\psi|0\rangle$:

$$\langle\psi|0\rangle = \left(\cos\left(\frac{\theta}{2}\right) \langle 0| + e^{-i\phi} \sin\left(\frac{\theta}{2}\right) \langle 1| \right) |0\rangle \quad (13)$$

$$= \cos\left(\frac{\theta}{2}\right) \quad (14)$$

Therefore:

$$|\langle\psi|0\rangle|^2 = \cos^2\left(\frac{\theta}{2}\right) \quad (15)$$

$$F(\psi) = \frac{1}{2} + \frac{1}{2} \cos^2\left(\frac{\theta}{2}\right) \quad (16)$$

$$= \frac{1}{2} \left(1 + \cos^2\left(\frac{\theta}{2}\right) \right) \quad (17)$$

$$\boxed{F(\psi) = \frac{1}{2} \left(1 + \cos^2\left(\frac{\theta}{2}\right) \right)} \quad (18)$$

4 Question 2: Is the Cloner Symmetric?

4.1 Definition of Symmetric Cloner

A cloner is symmetric if both output qubits have the same reduced density matrix:

$$\rho_1 = \rho_2 \quad (19)$$

4.2 Reduced Density Matrix of Second Qubit

Tracing out the first qubit:

$$\rho_2 = \text{Tr}_1[\rho_{\text{out}}] \quad (20)$$

$$= \text{Tr}_1 \left[\frac{1}{2} |\psi\rangle \langle \psi| \otimes |0\rangle \langle 0| \right] + \text{Tr}_1 \left[\frac{1}{2} |0\rangle \langle 0| \otimes |\psi\rangle \langle \psi| \right] \quad (21)$$

$$= \frac{1}{2} |0\rangle \langle 0| \cdot \text{Tr}[|\psi\rangle \langle \psi|] + \frac{1}{2} |\psi\rangle \langle \psi| \cdot \text{Tr}[|0\rangle \langle 0|] \quad (22)$$

$$= \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |\psi\rangle \langle \psi| \quad (23)$$

4.3 Comparison

We have:

$$\rho_1 = \frac{1}{2} |\psi\rangle \langle \psi| + \frac{1}{2} |0\rangle \langle 0| \quad (24)$$

$$\rho_2 = \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |\psi\rangle \langle \psi| \quad (25)$$

Since addition is commutative:

$$\rho_1 = \rho_2 \quad (26)$$

Yes, the cloner is symmetric.

(27)

5 Question 3: Is the Cloner Universal?

5.1 Definition of Universal Cloner

A cloner is universal if the fidelity $F(\psi)$ is independent of the input state $|\psi\rangle$.

5.2 Analysis

From our result in Question 1:

$$F(\psi) = \frac{1}{2} \left(1 + \cos^2 \left(\frac{\theta}{2} \right) \right) \quad (28)$$

This fidelity depends on θ (but not on ϕ). Different input states with different values of θ will have different fidelities.

For example:

- For $|\psi\rangle = |0\rangle$ (i.e., $\theta = 0$):

$$F(|0\rangle) = \frac{1}{2} (1 + \cos^2(0)) = \frac{1}{2} (1 + 1) = 1 \quad (29)$$

- For $|\psi\rangle = |1\rangle$ (i.e., $\theta = \pi$):

$$F(|1\rangle) = \frac{1}{2} (1 + \cos^2(\pi/2)) = \frac{1}{2} (1 + 0) = \frac{1}{2} \quad (30)$$

- For $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ (i.e., $\theta = \pi/2$):

$$F = \frac{1}{2}(1 + \cos^2(\pi/4)) = \frac{1}{2}(1 + 1/2) = \frac{3}{4} \quad (31)$$

Since $F(\psi)$ varies with θ :

$$\boxed{\text{No, the cloner is not universal.}} \quad (32)$$

6 Question 4: Average Single-Copy Fidelity

6.1 Definition

The average fidelity is obtained by averaging over all possible input states uniformly distributed on the Bloch sphere:

$$F_{\text{average}} = \int \frac{d\Omega}{4\pi} F(\psi) \quad (33)$$

where $d\Omega = \sin \theta d\theta d\phi$ is the solid angle element.

6.2 Computation

$$F_{\text{average}} = \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta F(\psi) \quad (34)$$

$$= \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \frac{1}{2} \left(1 + \cos^2 \left(\frac{\theta}{2} \right) \right) \quad (35)$$

Since the integrand doesn't depend on ϕ :

$$F_{\text{average}} = \frac{2\pi}{4\pi} \int_0^\pi \sin \theta d\theta \frac{1}{2} \left(1 + \cos^2 \left(\frac{\theta}{2} \right) \right) \quad (36)$$

$$= \frac{1}{4} \int_0^\pi \sin \theta \left(1 + \cos^2 \left(\frac{\theta}{2} \right) \right) d\theta \quad (37)$$

Let's compute each integral separately.

6.2.1 First Integral

$$I_1 = \int_0^\pi \sin \theta d\theta = [-\cos \theta]_0^\pi = -(-1) - (-1) = 2 \quad (38)$$

6.2.2 Second Integral

$$I_2 = \int_0^\pi \sin \theta \cos^2 \left(\frac{\theta}{2} \right) d\theta \quad (39)$$

Use the substitution $u = \frac{\theta}{2}$, so $du = \frac{1}{2}d\theta$ and $\sin \theta = 2 \sin(u) \cos(u)$:

$$I_2 = \int_0^{\pi/2} 2 \sin(u) \cos(u) \cos^2(u) \cdot 2 du \quad (40)$$

$$= 4 \int_0^{\pi/2} \sin(u) \cos^3(u) du \quad (41)$$

Let $v = \cos(u)$, so $dv = -\sin(u)du$:

$$I_2 = -4 \int_1^0 v^3 dv = 4 \int_0^1 v^3 dv \quad (42)$$

$$= 4 \left[\frac{v^4}{4} \right]_0^1 = 4 \cdot \frac{1}{4} = 1 \quad (43)$$

6.2.3 Final Calculation

$$F_{\text{average}} = \frac{1}{4}(I_1 + I_2) \quad (44)$$

$$= \frac{1}{4}(2 + 1) \quad (45)$$

$$= \frac{3}{4} \quad (46)$$

$$\boxed{F_{\text{average}} = \frac{3}{4}} \quad (47)$$

7 Question 5: Comparison with Bužek-Hillery Cloner

7.1 Bužek-Hillery Optimal Symmetric Universal Quantum Cloner

The Bužek-Hillery optimal symmetric universal quantum cloning machine for $1 \rightarrow 2$ cloning of arbitrary qubit pure states achieves an average single-copy fidelity of:

$$F_{\text{BH}} = \frac{5}{6} \approx 0.833 \quad (48)$$

7.2 Comparison

Our cloner achieves:

$$F_{\text{average}} = \frac{3}{4} = 0.75 \quad (49)$$

$$F_{\text{BH}} - F_{\text{average}} = \frac{5}{6} - \frac{3}{4} \quad (50)$$

$$= \frac{10}{12} - \frac{9}{12} \quad (51)$$

$$= \frac{1}{12} \approx 0.083 \quad (52)$$

$$\boxed{F_{\text{average}} = \frac{3}{4} < F_{\text{BH}} = \frac{5}{6}} \quad (53)$$

7.3 Interpretation

The average single-copy fidelity of our cloner is **lower** than the Bužek-Hillery optimal symmetric universal quantum cloning machine by $\frac{1}{12}$.

This makes sense because:

- Our cloner is symmetric (both copies have the same quality)
- However, it is **not** universal (fidelity depends on input state)
- The Bužek-Hillery cloner is both symmetric **and** universal
- The Bužek-Hillery cloner is optimal among all symmetric universal cloners

Our cloner performs better for states close to $|0\rangle$ (achieving $F = 1$ for $|0\rangle$) but worse for states in the equatorial plane of the Bloch sphere.

8 Summary of Answers

1. Single-copy fidelity:

$$F(\psi) = \frac{1}{2} \left(1 + \cos^2 \left(\frac{\theta}{2} \right) \right) \quad (54)$$

2. Is the cloner symmetric?

$$\text{Yes, } \rho_1 = \rho_2 \quad (55)$$

3. Is the cloner universal?

$$\text{No, } F(\psi) \text{ depends on } \theta \quad (56)$$

4. Average single-copy fidelity:

$$F_{\text{average}} = \frac{3}{4} \quad (57)$$

5. Comparison with Bužek-Hillery:

$$F_{\text{average}} = \frac{3}{4} < F_{\text{BH}} = \frac{5}{6} \quad (58)$$

The cloner performs worse than the optimal Bužek-Hillery cloner by $\frac{1}{12} \approx 8.3\%$ on average.